

Purple Team Outline

Week 06: Scripting & Automation in Network Security

Objective: To master the use of higher-level languages for automating repetitive security tasks and building custom Purple Team tools..

Task : Automation Fundamentals & Environmental Setup With Practical Security Scripting (Python & Bash)

- **Description:** Understand the role of automation in scaling both offensive operations and defensive monitoring & create functional scripts to automate network reconnaissance and log parsing.
- **Technical Requirements:**
 - Define the importance of scripting in modern SOC environments and Red Team engagements.
 - Set up a Python and Bash development environment within the lab.
 - Analyze sample automation tools and identify the libraries used (e.g., Scapy, Requests, or Subprocess).
 - **Bash:** Create a script to automate the discovery of live hosts and open ports using local system tools.
 - **Python:** Develop a script to parse a web server log file and extract all unique IP addresses or identify potential SQLi attempts.
 - **Integration:** Use Python's Scapy library to crafter for testing IDS signatures.
- **Learning Goal:** Recognize how automation reduces human error and increases the speed of response/attack & Gain the ability to "build your own tools" instead of relying solely on pre-built software.

- **What to Document :**

- **Day 1:**

- What is **automation in cybersecurity** & Role of scripting in **SOC operations**.
 - Role of scripting in **Red Team engagements**.
 - Identify 3 security tools that rely on scripting internally.
 - Network scanning tools
 - Log analysis tools
 - Packet crafting tools
 - For each tool write:
 - Purpose
 - Programming language used
 - Automation capability

- **Day 2:**

- Verify installations:
 - python3 --version
 - bash --version
 - Install required libraries:
 - pip install scapy
 - pip install requests
 - Research the following Python libraries:
 - Write a short explanation of **how each library could help security automation**.

- **Day 3:**

- Analyze any **two automation tools**:
 - Nmap
 - Wireshark
 - Metasploit
 - Identify:
 - What tasks they automate
 - What programming languages are used internally
 - Which libraries might be involved
 - Create a **simple automation workflow map** & Explain **which steps could be automated with Bash or Python**:
 - Scan network → Identify live hosts → Detect open ports → Log results

○ **Day 4:**

- Create a Bash script that:
 - Accepts a subnet as input
 - Pings hosts to identify **live systems**
 - Runs a port scan using system tools
 - **Saves results to a file**
- Example tools allowed:
 - Ping
 - Netcat
 - Nmap
 - Ss
 - Grep
- Example Logic:

```
Input subnet
For each host in range
  Ping host
  If alive
    Scan ports
    Save results
```

- Expected Output:

```
Live Host: 192.168.1.5
Open Ports: 22,80
```

○ **Day 5:**

- Write a Python script that:
 - Reads an **Apache/Nginx log file**
 - Extracts **unique IP addresses**
 - **Detects potential SQL Injection attempts**
- Look for patterns such as:

```
' OR '1'='1
UNION SELECT
--
DROP TABLE
```

- Script Output Example:

```
Unique IPs Found: 12
Suspicious Requests:
192.168.1.10 → ' OR '1'='1
```

- **Day 6:**

- Create a Python script that:
 - Uses Scapy to craft packets
 - Sends packets to a test machine
 - **Simulates suspicious traffic**
- Example packet types:
 - ICMP echo requests
 - TCP SYN packets
 - Custom payload packets

- Example concept:

```
from scapy.all import *  
packet = IP(dst="192.168.1.5")/ICMP()  
send(packet)
```

- How did automation reduce manual work in your scripts?
- What risks exist when automating offensive security tools?
- How could these scripts help a SOC analyst respond faster?